Emotion Detection API Documentation

Overview

The **Voice Compassion API** is a FastAPI-based service that detects emotions from audio and video files using OpenAI Whisper large-v3 model deployed on Triton Inference Server. The API automatically extracts audio from video files and performs emotion classification.

Version: 1.0.0
Base URL: https://se-api-dgx.nimar.gov.pk

Authentication

All protected endpoints require Bearer token authentication.

Header Format:

```
Authorization: Bearer <API_KEY>
```

Default API Key: 123

Endpoints

1. Detect Emotion from Audio/Video

Endpoint: POST /v1/voice_compassion
Authentication: Required
Content-Type: multipart/form-data

Uploads an audio or video file and returns the detected emotion.

Request Parameters

Parameter	Type	Required	Description
video_file	file	Yes	Audio or video file to analyze

Supported File Formats

Audio Formats:

- .wav, .mp3, .flac, .aac, .ogg, .opus, .m4a, .aiff, .amr, .wma

Video Formats:

- .mp4, .mkv, .webm, .avi, .mov, .mpg, .mpeg, .flv, .3gp, .ts, .f4v

Response

Success (200):

```
{
  "emotion": "happy"
}
```

Possible Emotions:

- happy
- sad
- angry
- neutral
- fearful
- disgusted
- surprised

Error Responses

401 Unauthorized - Missing Authorization:

```
{
  "detail": "Missing Authorization header"
}
```

401 Unauthorized - Invalid API Key:

```
```

```
{
  "detail": "Invalid API key"
}
```

400 Bad Request - No Audio in Video:

```json
{
  "error": "The provided file does not contain any audio"
}
```

503 Service Unavailable:

```json
{
  "detail": "Triton inference service is not available"
}
```

cURL Example

```bash
curl -X POST "https://se-api-dgx.nimar.gov.pk/v1/voice_compassion" \
  -H "Authorization: Bearer 123" \
  -F "video_file=@/path/to/audio.wav"
```

Python Example

```python
```

```python
import requests

url = "https://se-api-dgx.nimar.gov.pk/v1/voice_compassion"
headers = {
    "Authorization": "Bearer 123"
}
files = {
    "video_file": open("audio.wav", "rb")
}

response = requests.post(url, headers=headers, files=files)
print(response.json())
```

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2. Health Check

Endpoint: `GET /health`
Authentication: Not Required

Check the health status of the API and Triton inference server.

Response

Success (200):

```json
{
  "status": "healthy",
  "timestamp": "2025-11-05T10:30:00.123456",
  "version": "1.0.0"
}
```

Service Unavailable (503):

```json
```

```
{
  "detail": {
    "status": "unhealthy",
    "timestamp": "2025-11-05T10:30:00.123456",
    "version": "1.0.0"
  }
}
```

cURL Example

```bash
curl -X GET "https://se-api-dgx.nimar.gov.pk/health"
```

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3. Get Configuration

Endpoint: `GET /config`
Authentication: Not Required

Retrieve current API configuration including Triton server details and supported formats.

Response

```json
{
  "triton_server_url": "192.168.18.22:32216",
  "model_name": "speech_emotion_recognition",
  "supported_extensions": [
    ".wav", ".mp3", ".flac", ".aac", ".ogg", ".opus", ".m4a",
    ".mp4", ".mkv", ".webm", ".avi", ".mov", ".mpg"
  ]
}
```

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4. API Information

Endpoint: `GET /`
Authentication: Not Required

Get general API information and available endpoints.

Response

```json
{
  "service": "Voice Compassion API",
  "version": "1.0.0",
  "description": "Voice Compassion API using Triton Inference Server",
  "endpoints": {
    "voice_compassion": "/v1/voice-compassion/",
    "health": "/health",
    "config": "/config",
    "docs": "/docs"
  },
  "supported_formats": {
    "video": [".mp4", ".mkv", ".webm", ".avi", ".mov", ".mpg", ".mpeg", ".flv", ".3gp", ".ts", ".f4v"],
    "audio": [".wav", ".mp3", ".flac", ".aac", ".ogg", ".opus", ".m4a", ".aiff", ".amr", ".wma"]
  }
}
```

Interactive Documentation

Access interactive API documentation at:

- **Swagger UI:** https://se-api-dgx.nimar.gov.pk/docs
- **ReDoc:** https://se-api-dgx.nimar.gov.pk/redoc

Rate Limits & Performance

- **Max File Size:** 500MB (configurable via Ingress)
- **Request Timeout:** 600 seconds
- **Auto-scaling:** 2-10 replicas based on load
- **Workers per Pod:** 4 uvicorn workers

Error Handling

All errors follow standard HTTP status codes:

Status Code	Description
200	Success
400	Bad Request (invalid file, no audio, etc.)
401	Unauthorized (missing/invalid API key)
422	Validation Error (missing required fields)
500	Internal Server Error
503	Service Unavailable (Triton server down)

Best Practices

1. **File Size:** Keep files under 100MB for optimal performance
2. **Audio Quality:** Use 16kHz sample rate for best results
3. **Video Processing:** Expect longer processing times for video files
4. **Error Handling:** Always implement retry logic for 503 errors
5. **Authentication:** Store API keys securely in environment variables

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Testing with Postman

1. Import the provided Postman collection
2. Update the `BASE_URL` variable if using a different domain
3. Update the `API_KEY` variable with your actual API key
4. Select a sample audio/video file for testing
5. Send the request to `/v1/voice_compassion`

Kubernetes Deployment

The API is deployed on Kubernetes with:

- **Namespace:** `emotion-detection-api`
- **Replicas:** 2 (min) to 10 (max) with HPA
- **Resources:** 1-4GB RAM, 0.5-2 CPU per pod

- **Ingress:** NGINX with 500MB max body size

Support & Troubleshooting

Common Issues

1. "Triton inference service is not available"

- Check if Triton server is running at `192.168.18.22:32216`
- Verify network connectivity between API pods and Triton server

2. "The provided file does not contain any audio"

- Ensure video file has an audio track
- Try converting video with FFmpeg: `ffmpeg -i input.mp4 -vn output.wav`

3. Timeout errors

- For large files, the processing may take longer
- Check the request timeout settings (default: 600s)

Model Information

Model: `firdhokk/speech-emotion-recognition-with-openai-whisper-large-v3`
Backend: Triton Inference Server
Sample Rate: 16kHz
Task Type: Emotion Classification

Contact

For technical support or API access requests, contact the MLOps team at NIMAR.

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